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# UK Election 2024: Forecast Mapping

This page explores some visualizations relating to the 4th July UK General Election. It serves to explore

- the predicted outcomes of the general election
- comparison of the predictions with the actual results
- the variation in different approaches to result prediction
- different visualization designs, especially those that use [hex cartograms](#) to represent constituencies.

## Election Data Vault Forecasts

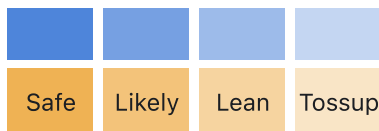
[Election data vault](#) have been archiving UK election opinion polls and election results since 1945. In the run-up to the 2024 General Election, they provide [nowcasts](#) and [forecasts](#) of the predicted result based on an aggregation of the most recent opinion polling.

### Comparing 2019 results with 2024 forecast

Based on the **30th of June forecast**, four days before the actual election, the maps below compare the real results from 2019 and 2024 with Election Vault's final polling prediction before the 2024 election. Bolder colours in the forecast map indicate greater certainty in the predicted result. Below each map the constituencies are ordered by total number of majorities for each party, signifying the actual or predicted makeup of the Westminster parliament.

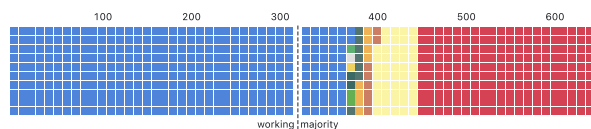
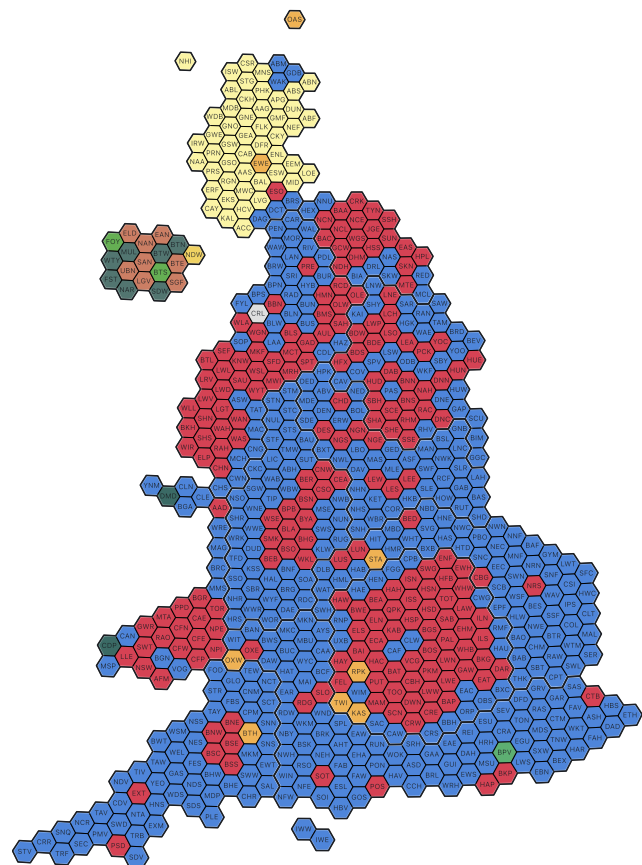
Predictions carry a degree of uncertainty which is classified according to the proportion of simulations that yield a given result. Certainty of prediction is symbolised by the strength of colour:





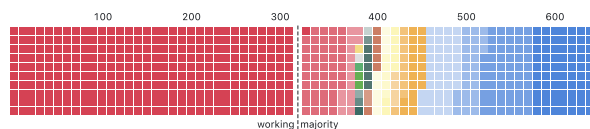
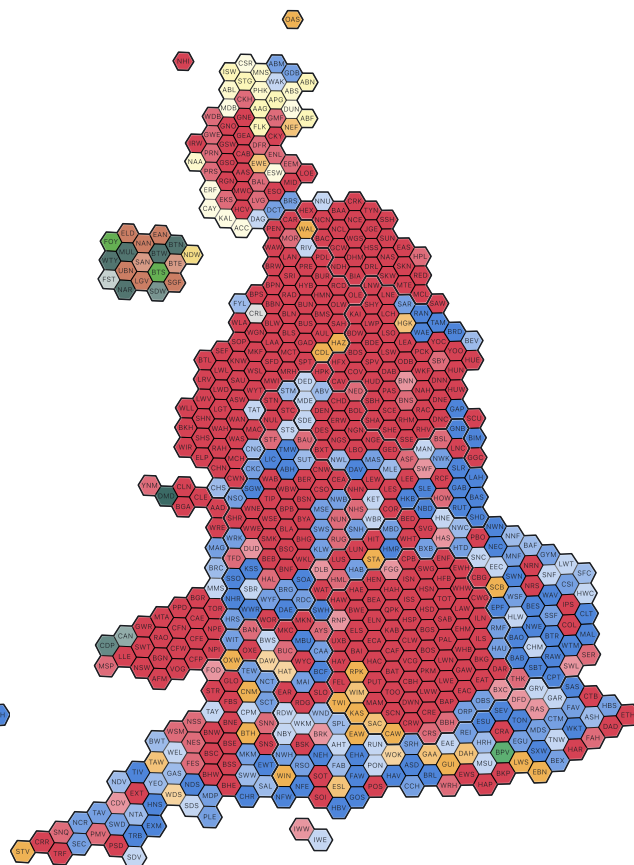
## 2019 General Election

Estimated projection onto 2024 constituencies



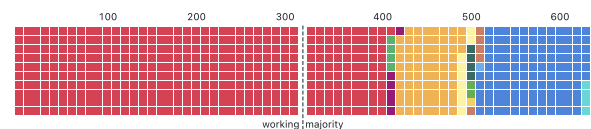
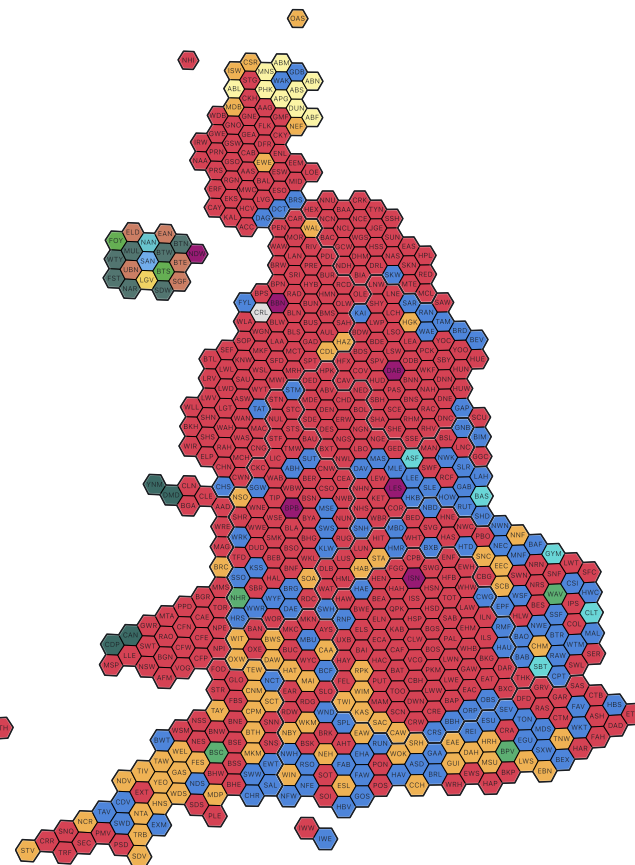
## 2024 General Election forecast

Based on 30 June 2024 polling prediction



## 2024 General Election

Confirmed results



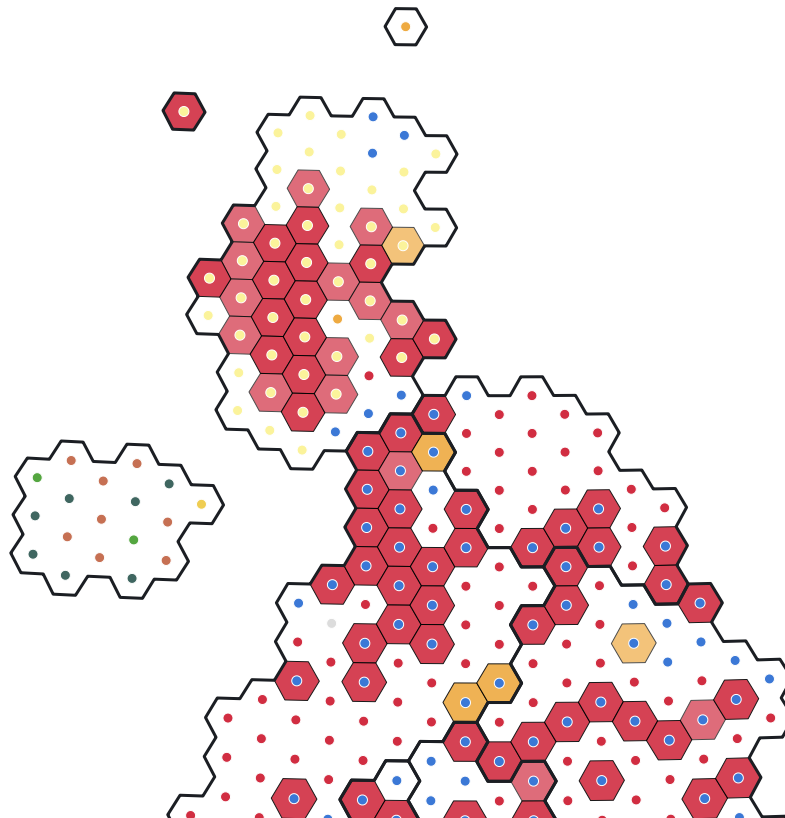
To examine where the change in voting preference is predicted to change the party of the elected MP, below the map only colours hexes for constituencies that are predicted to change. The small dots show the 2019 result for comparison.

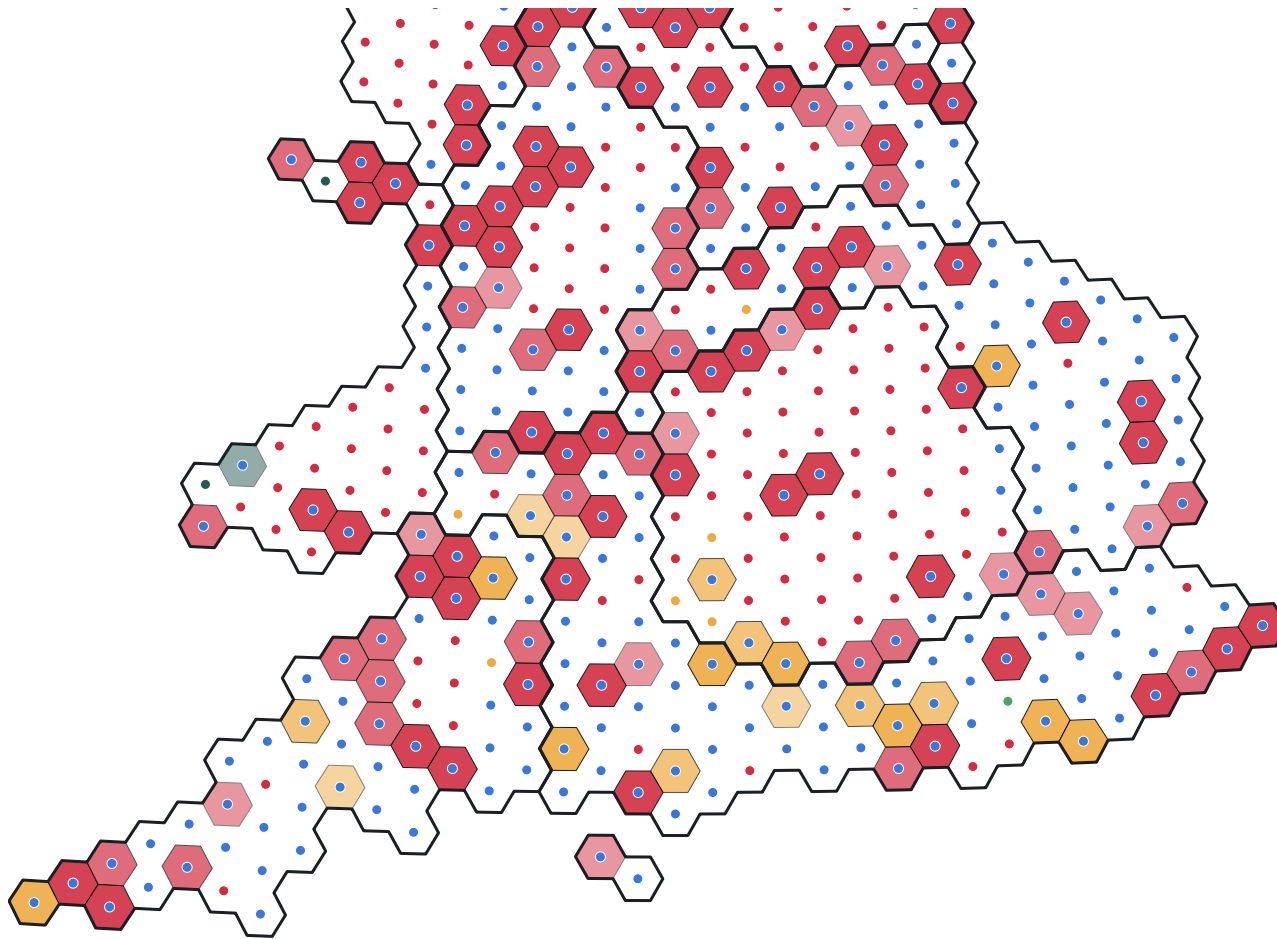
This makes it easier to see the so-called 'red wall' (largely in the Midlands and north of England) that is likely to revert back to Labour and the smaller 'blue wall' (largely in the south of England) that is likely to see shift from Conservative to Liberal Democrats and Labour. It also highlights the big change in Scotland with a collapse in the popularity of the SNP in favour of the Labour Party.

## Predicted change in winning party 2019 to 2024

Small dot: 2019 result (no change); Hex: seat with a predicted change in 2024

Con Lab LD Grn SNP PC DUP SF All SDLP Speaker



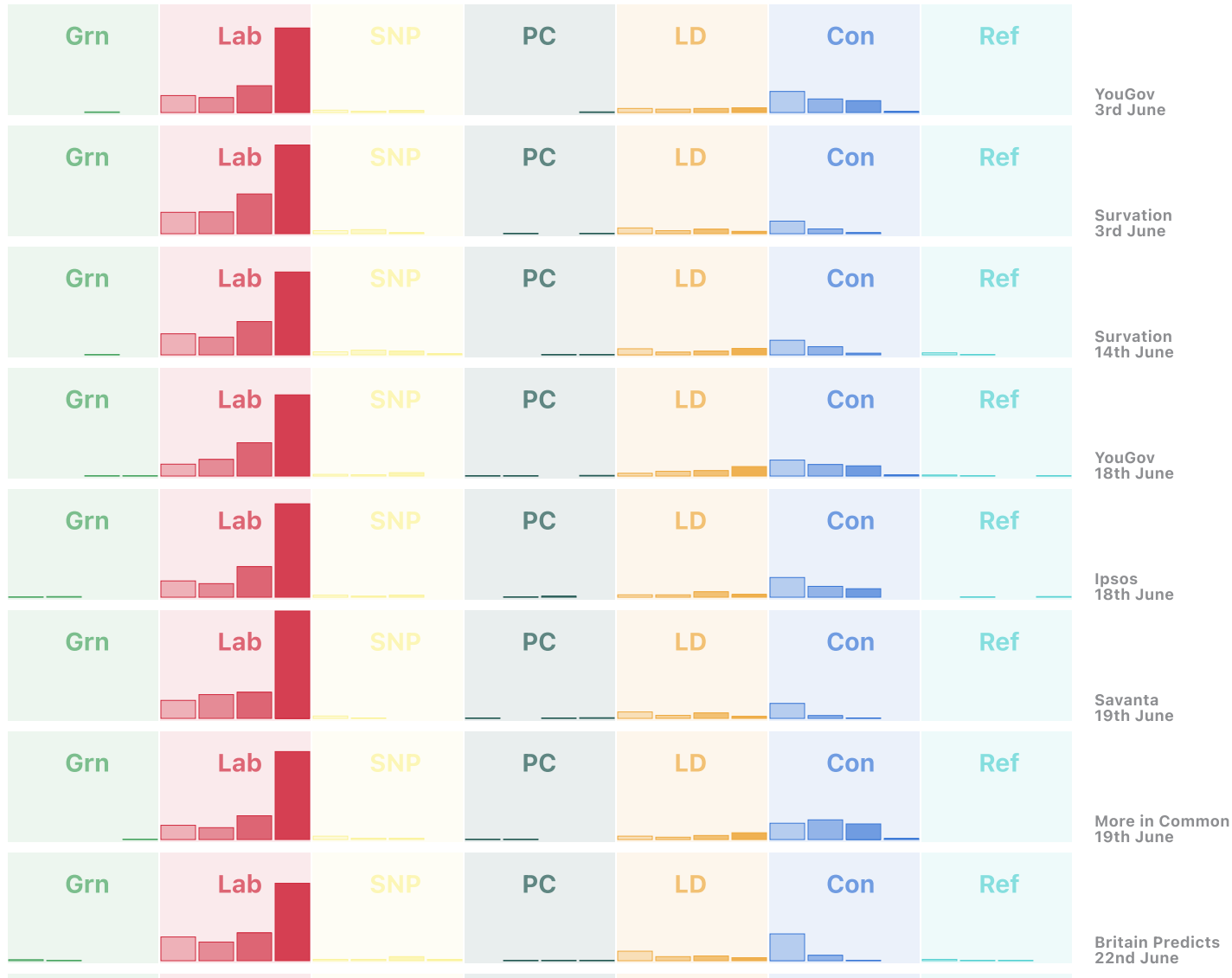


## MRP Modelling

A recent trend has seen polling companies use [multilevel regression with poststratification](#) (MRP) in an attempt to improve the accuracy of predictions from polling returns. It attempts to provide a more realistic mapping of opinion poll response to actual voting behaviour by using the demographic and behavioural characteristics (e.g. propensity to actually vote on election day) to weight sample responses by local area. The result is a constituency level set of predicted voting behaviours. Note that the MRPs shown here do not include Northern Ireland or the current (non-voting) Speaker's constituency of Chorley.

This can lead to quite different predictions between polling companies depending on how those demographic and behavioural characteristics are measured, as well as the more conventional problems of sample sizes and representativeness.

Here are the distributions of most likely results from the MRPs so far released along with their average predictions. As above, the strength of colour indicates more certain outcomes, as measured by the likely majority.



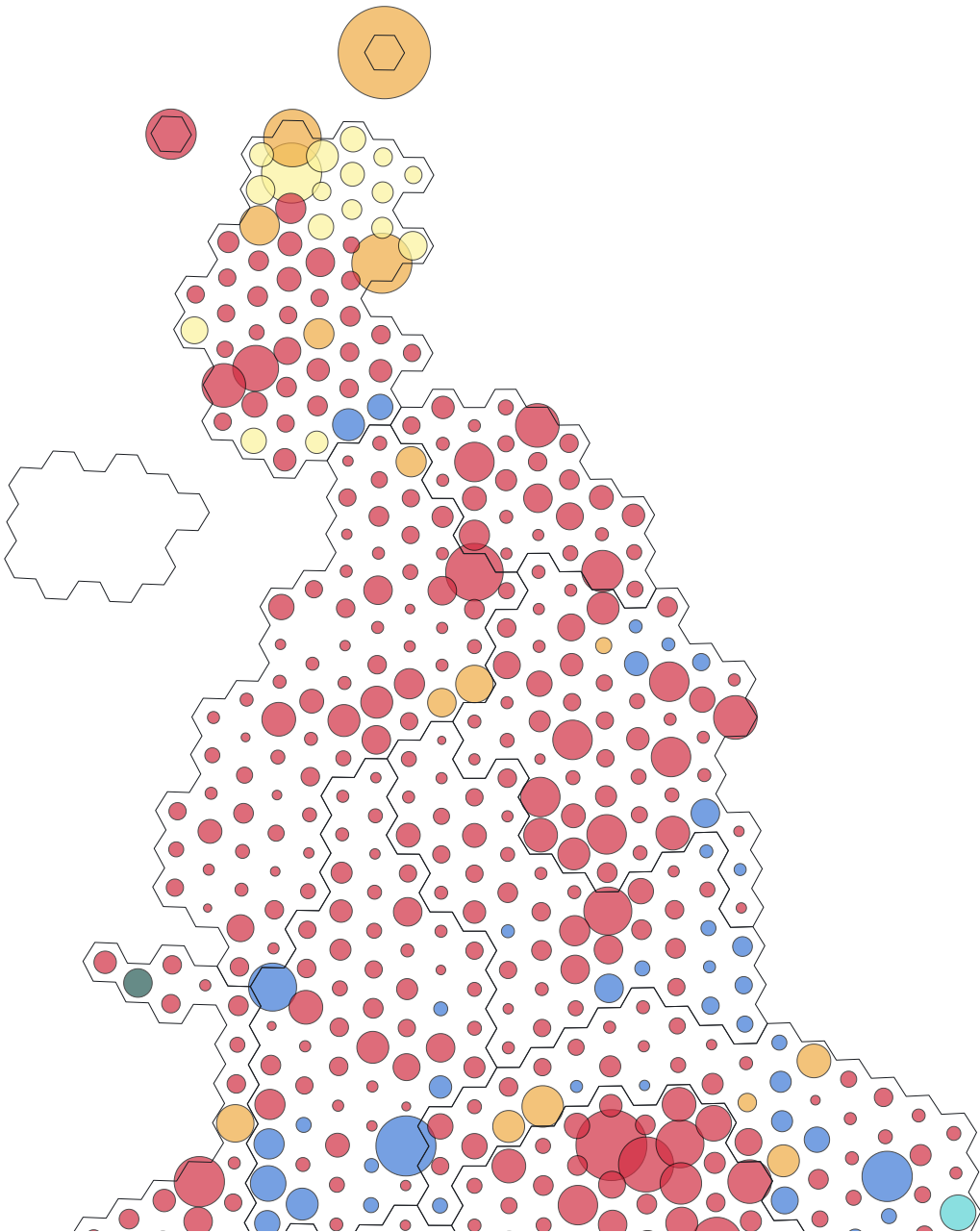


Where do the MRPs agree and where do they differ the most? The map below shows how varied the predicted constituency results are between the full set of MRPs. The larger the circle, the greater the difference in prediction and therefore the more uncertain the result.

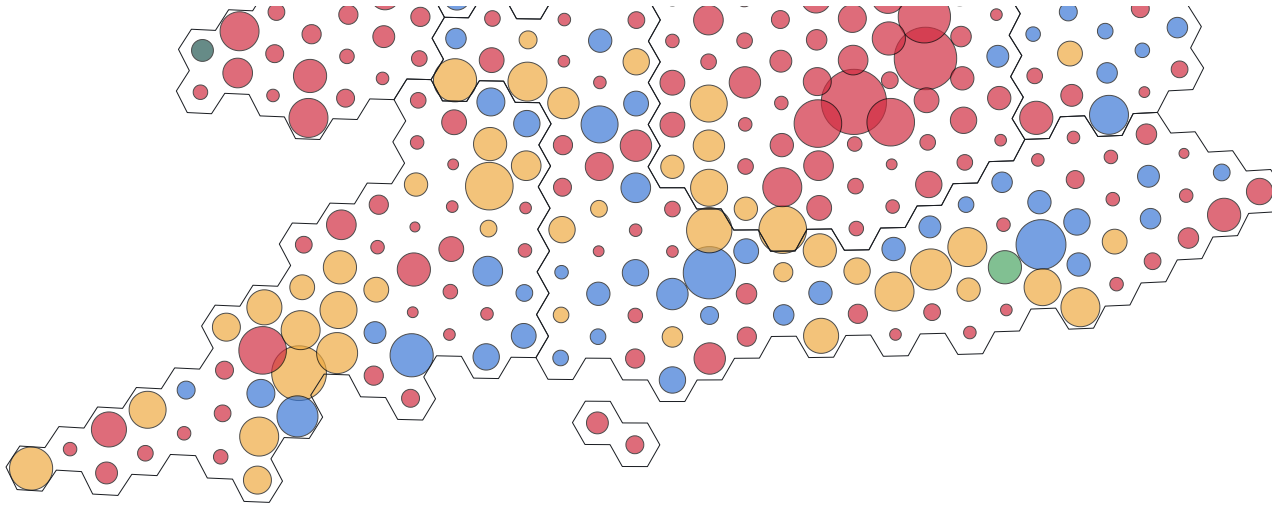
# Variation in MRP predictions

Average standard deviation of predicted party vote by constituency

Con Lab LD Grn Ref SNP PC





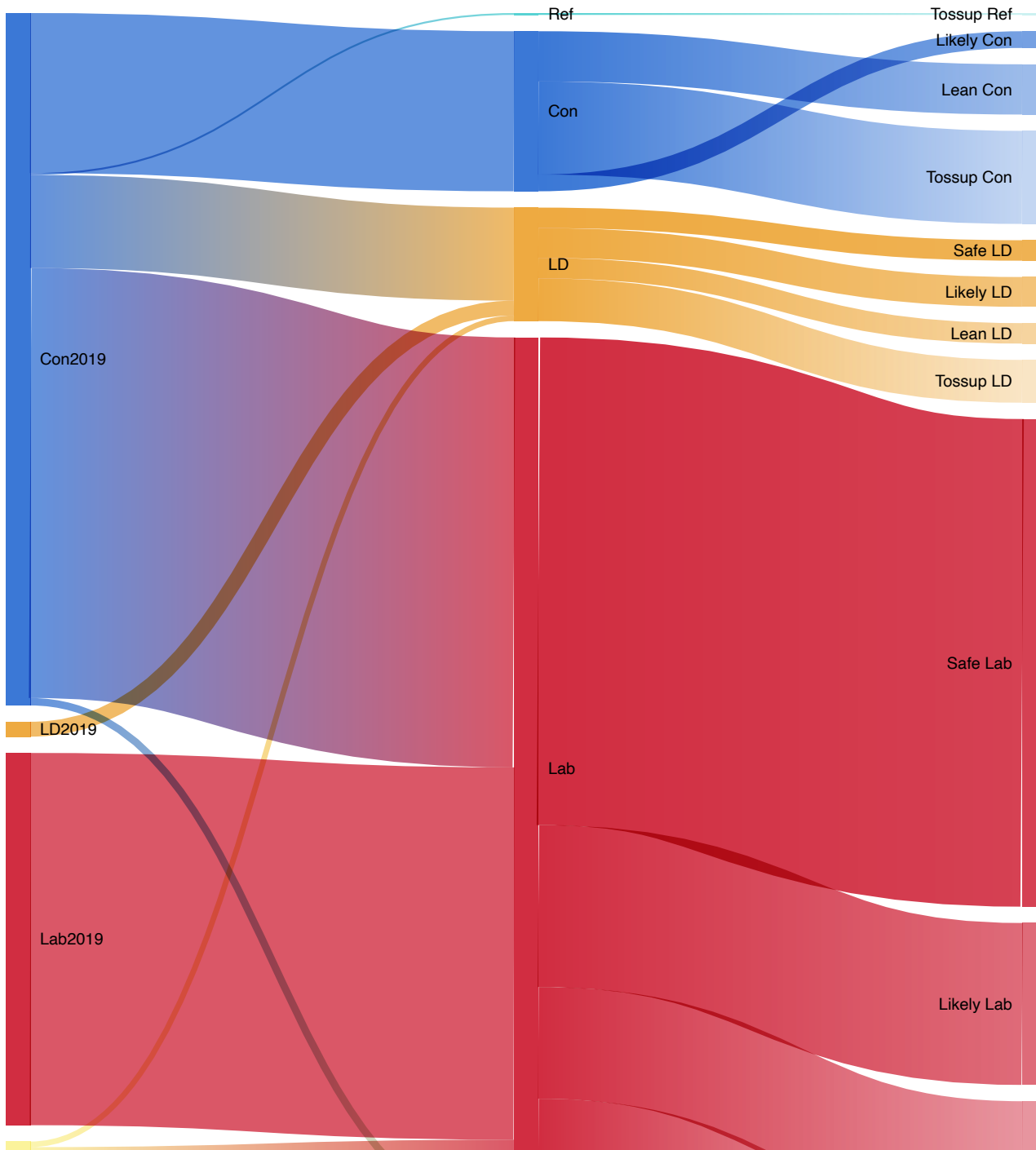


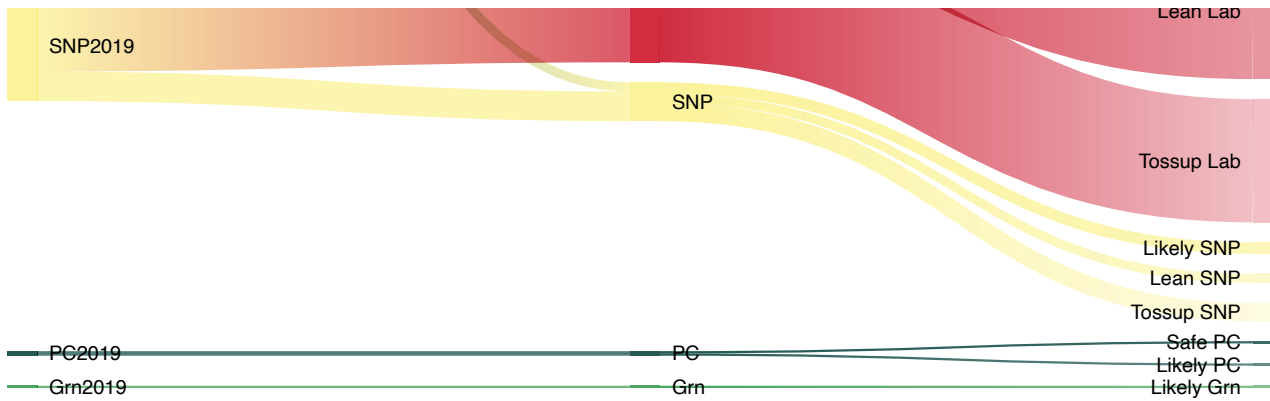
The MRPs have some trouble in predicting Scottish seats where there is a close call between the SNP and others. This is in part because of the unpredictability of pro/anti union tactical voting.

London has some uncertainty in areas that have traditionally be Labour supporting but where there has been concern about Gaza and rhetoric about immigration from the Labour Party. It is not clear the extent to which this will transform into changed voting behaviour.

Constituencies with specific local circumstances also show larger variation between MRPs. For example, the four Green Party target seats (Brighton Pavilion, Waveney Valley, Bristol Central and North Herefordshire) and high profile independents (Jeremy Corbyn in Islington North; Jason Zadrozny in Ashfield; Nigel Farage in Clacton); complex local circumstances (Blaenau Gwent with independent Labour and vote of no confidence in the incumbent Labour MP; North Shropshire with by-election win for the LibDems in a traditional Conservative constituency after resignation of Owen Paterson).

We can compare the various MRPs (right) with the 2019 election result (left) to show how seats are predicted to change and in which direction:





Caution should be exercised in interpreting the changes. While it may be tempting to ascribe them to individuals (a former Conservative voter changes their vote to Labour), these are aggregated data by constituency. For example, an apparent switch from Conservative to Labour could be produced by former Conservatives not voting and former non voters voting Labour.

To see the geographic distribution of different MRPs select from the two lists below:

MRP 1

YouGov, July 3

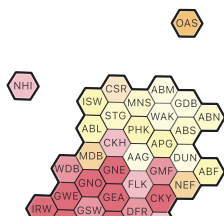
Hex map ☒

MRP 2

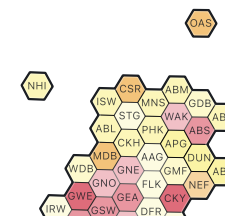
YouGov, July 3

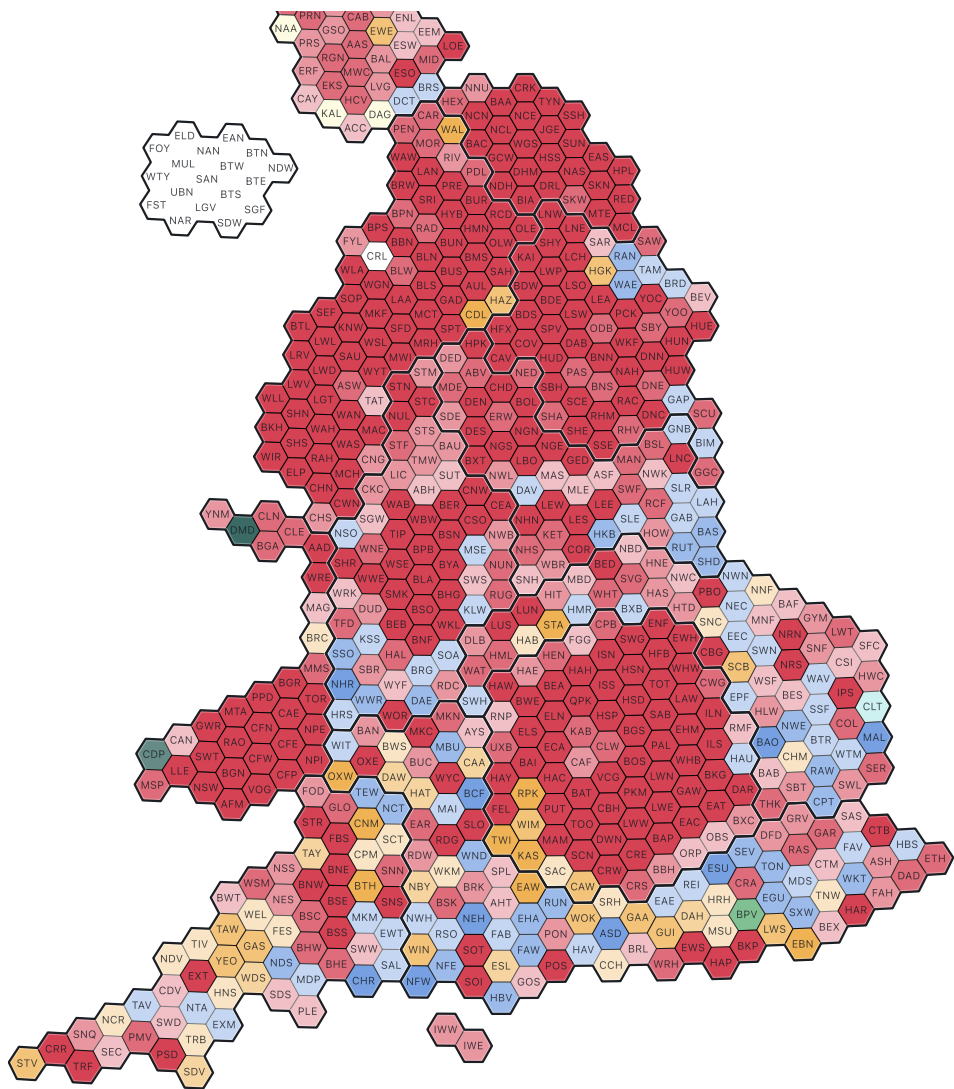
Hex map ☒

## Average of MRPs



## We Think (Economist), Jun 26





100

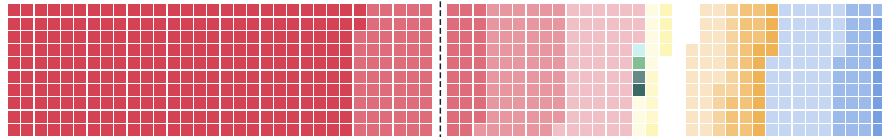
200

300

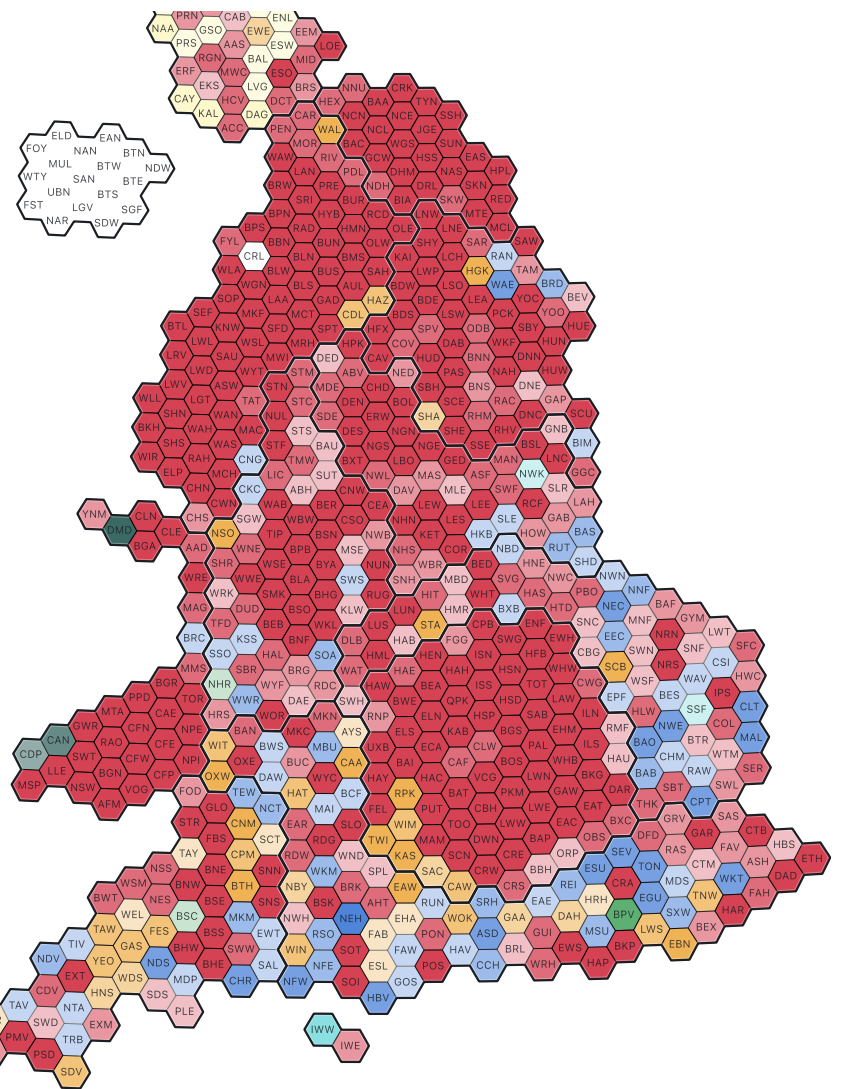
400

500

600



working | majority



100

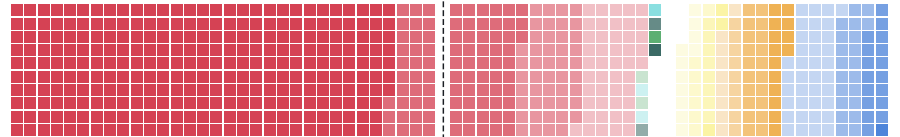
200

300

400

500

600



working | majority

# Observations

## Late June MRPs

FocalData released [their MRP](#) on the 24th June and two days later, MRPs from [We Think for the Economist](#) and [Find Out Now for Electoral Calculus](#) were released.

Find Out Now (Electoral Calculus) is unusual in the MRPs posted so far in that it is much more favourable to the smaller parties and a disaster for the Conservatives with the Liberal Democrats predicted as the official opposition and 18 seats to Reform. It also gives all 4 of the Greens' target seats to them and Jeremy Corbyn as an independent in Islington North. The MRP is very much an outlier, but will be interesting to see how right or wrong it ends up being.

## Early June MRPs

Two MRP results were published on June 3 – one from [YouGov](#) and the other from [Survation](#).

Several striking observations can be made:

- The two MRP results differ quite significantly in the wide band of constituencies that are not strongly leaning to a one party. Geographically, this can be seen in the variation of the southern England, non-London constituencies.
- Despite the differences, both are clearly showing a very large Labour parliamentary majority.
- *There are no projected 'safe' conservative seats.* Even the most strongly Conservative fall into the 'likely' rather than 'safe' categories based on the probability of a party majority.

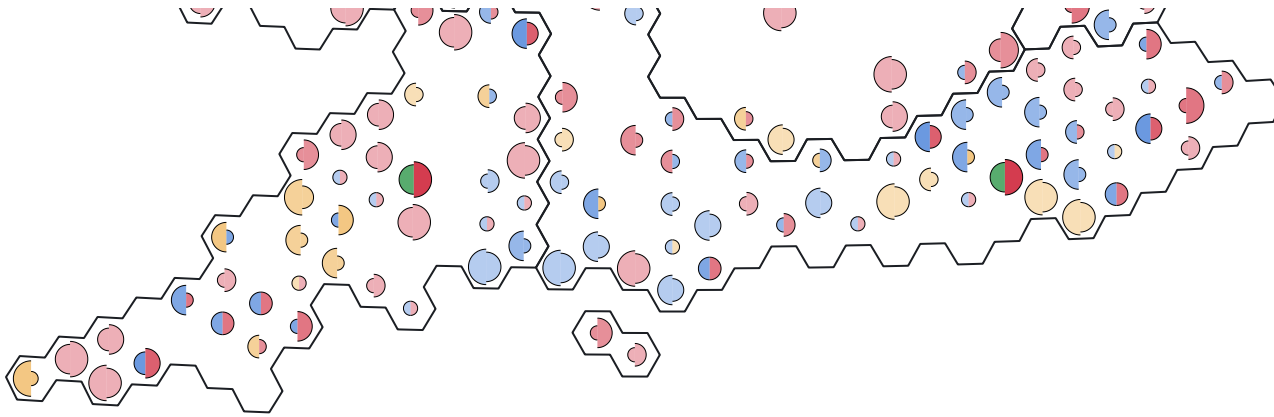
How different are the two MRP models? Here are the constituencies with a difference in predicted result. Greater differences in the two MRPs are shown in bolder colours. No symbols are shown where the two models agree (generally in safe seats).

## Difference in seat predictions in YouGov and Survation MRPs

Left semicircle: YouGov; Right semicircle: Survation; Size: size of predicted majority

Con Lab LD Grn SNP PC





The interesting seats are where the two models predict different parties winning, especially if by a larger margin.

- The most extreme differences occur in the two seats (Bristol Central and Brighton Pavilion) where YouGov suggest a likely win for the Greens, but Survation suggest a safe Labour win. It highlights the challenge in using nationally gathered data to predict exceptional, but strong local behaviour.
- In Scotland there are some large differences in seats where the SNP vote may drop considerably.
- In eastern England and parts of the southwest England, there are many seats that differ in Labour-Conservative predicted winners, with YouGov's model generally favouring the more right-leaning result of the two.

---

## Appendices

*Boundary data and 2019 results projected to 2024 constituencies:*

```
import {geoData, geoRegions, hexData, hexRegions, notionalResults2019, partyClrs, partyPalette, fullPartyName, toLookup} from "@jwolondon/uk-election-2024-boundary-data"
```







```
mrpLabels = f(d)
```

*For a given list of MRPs, store the distribution of results:*

```
buildDistrib = f(mrps)
```

*Provide an opacity value for 4 certainty levels of each prediction:*

```
toCertainty = ► Map(5) {"Tossup" => 0.3, "Toss-Up" => 0.3, "Lean" => 0.5, "Likely" => 0.7, "Safe" => 0.9}
```

```
partyCertaintyClrs = ► Map(114) {"Con2019" => "rgb(59,120,214)", "Con" => "rgb(59,120,214)", "Safe Con" => "rgba(59,120,214,0.9)", "Tossup" => "rgba(59,120,214,0.3)", "Toss-Up" => "rgba(59,120,214,0.3)", "Lean" => "rgba(59,120,214,0.5)", "Likely" => "rgba(59,120,214,0.7)", "Safe" => "rgba(59,120,214,0.9)"}>
```

*Result and change mapping functions:*

```
partiesLR = ► Array(17) ["Grn", "WP", "Ind", "Lab", "LD", "SNP", "PC", "SDLP", "All", "DUP", "UUP", "other", "Ant", "Lib", "UKIP", "Brexit Party"]>
```

```
showResults = f(...)
```

```
waffle = f(...)
```

```
showAsWaffle = f(...)
```

```
certaintyLegend = f()
```

```
showChange = f(...)
```

```
showSimilarity = f(...)
```

```
showModelDifference = f(...)
```

```
showSD = f(...)
```

*Display distributions of several MRPs:*

*Display distributions of several mms.*

```
showDist = f(...)
```



*Create a custom semicircle symbol that behaves like Plot.dot, but with an additional direction: left/right option:*



```
semicircle = f(...)
```



*Given two poll results (winningPartyA, marginA, winningPartyB, marginB), quantify how different they are [0,1]:*

```
distance = f(pA, mA, pB, mB)
```

*Given two poll results (winningPartyA, marginA, winningPartyB, marginB), interpolate the steps between them:*

```
interpolate = f(pA, mA, pB, mB)
```

*Provide a sorting comparator that will sort an array of objects by the given keys (reverse order if prefixed with a -):*

```
sortRows = f(...)
```







